A Teacher-Guided Reinforcement Learning Pipeline for Role-Conditioned Football Cognition in Humanoid Robots

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***Abstract*—Humanoid robot soccer requires real-time tactical decisions under partial observability, limited communication, and strict onboard computation. This paper presents a teacher-guided reinforcement learning pipeline for role-conditioned high-level cognition on NAO V6 robots. Deterministic tactical heuristics generate validated demonstrations for approach, dribbling, shooting, passing, support, marking, blocking, and defensive recovery. Behavioral Cloning initializes a compact shared actor, which is subsequently fine-tuned through Proximal Policy Optimization in closed-loop SimRobot interaction. Behavioral-Cloning replay and Kullback–Leibler anchoring are incorporated to limit policy drift, although subsequent ablations reveal that removing KL improves held-out balanced accuracy from approximately 0.52 to 0.59. A deterministic coordinator assigns tactical roles, encoded within a 47-dimensional observation. The actor predicts eight skill logits and four bounded residual parameters, constrained by role-conditioned masks and action repair. Teacher-guided policies maintain higher skill accuracy and diversity than PPO alone, which predominantly selects walking. In 28 simulated 4v4 matches against B-Human, the final policy achieved 34.9% possession but recorded 7 wins, 4 draws, and 17 losses. These results support a compact, deployment-oriented hybrid architecture for learned humanoid soccer cognition while highlighting limitations in knowledge preservation and competitive performance.**

***Index Terms*—Humanoid robot systems, robot football, reinforcement learning, imitation learning, robot learning.**

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I. Introduction

Humanoid robot football is a highly complex decision-making problem where visual perception, locomotion, and tactical coordination intersect. Applying Reinforcement Learning in this domain extends beyond a robotic ball manipulation problem: it addresses operational challenges, such as multi-agent coordination [1], [2] and consistent sim-to-real transitions on physical hardware. Historically, robotic teams have relied on deterministic rule-based systems due to their safety and interpretability. However, those systems tend to have lower consistency as game interactions increase, creating a full range of conflicting contingencies. Recent studies demonstrate that learned policies handle continuous, noisy state spaces better [3], [4], and combining reinforcement-learned skills with hand-designed strategies has demonstrated competitive success in simulated humanoid robot soccer [5]. All the work reported here targets the NAO V6, the humanoid platform used by the team for football competition, whose limited onboard computation constrains the deployable policy size.

Rather than replacing deterministic cognition, the proposed platform reuses it as a structured teacher and runtime safety framework. Deterministic tactical heuristics generate

observation-action demonstrations, Behavioral Cloning transfers this expertise to a neural student [6], [7], and Proximal Policy Optimization refines the initialized policy through closed-loop interaction [8]. The article is organized as follows: Section II presents the case study and related work. Section III defines the reinforcement-learning formulation, environment, observation space, and action space. Section IV describes the teacher-guided training pipeline. Section V outlines the deployment sequence for the final ONNX agent, Section VI reports the experimental validation and discusses the results.

II. Case Study

Humanoid robot football requires decision making under noisy and incomplete observations, with limited communication and real-time constraints. Competitive systems separate team coordination, behavior selection, robot skills, and low-level motion control. B-Human follows this architecture, combining explicit tactics with reliable skills and real-time execution [9]; recent work shows coordinated play with restricted communication [10]. Dynamic role assignment has also been used to adapt team formations to the current game state [11]. Although reinforcement learning has produced agile soccer and locomotion skills [3], [7], the case study follows a hybrid approach in which learning is integrated into an existing robotics stack [12].

The proposed system controls three field players with one shared role-conditioned policy, while the goalkeeper uses a separate policy. A deterministic coordinator assigns striker, open-support, and off-ball-support roles from the general field state as ball specifics. Roles are therefore policy inputs rather than learned outputs. Previous robotic-soccer research has demonstrated the potential of multi-agent reinforcement learning for combining strategic decisions and control [1]. PPO-based methods have also provided strong results in cooperative multi-agent problems [8].The proposed architecture is more precisely characterized as centralized deterministic role assignment with decentralized shared-policy execution supported by team-state communication, rather than end-to-end learned coordination.

Starting reinforcement learning from a random policy produces many tactically unproductive interactions. Instead, a hand-designed multi-agent teacher generates demonstrations, and behavior cloning initializes the policy before RL fine-tuning. This refinement is necessary because imitation alone accumulates errors when the policy visits states outside the demonstration distribution, a problem known as covariate shift [13]. During fine-tuning, behavior-cloning replay and a KL penalty to a frozen reference policy limit abrupt degradation, in a manner related to kickstarting [14].

The deployed policy receives a 47-dimensional observation and produces eight skill logits, plus four continuous skill parameters as shown in Figure 3. Combining a discrete decision with continuous arguments follows the parameterized-action formulation previously applied to simulated robot football [15]. Before selecting a skill, a state-dependent mask

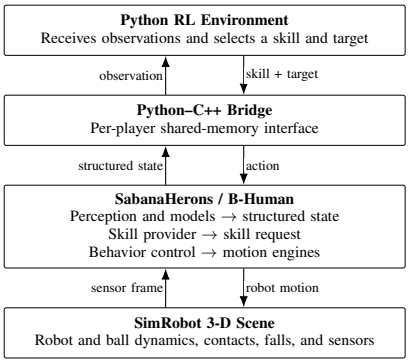


Fig. 1. SabanaHerons–SimRobot training loop.



Fig. 2. Match simulation using simrobot.

removes actions that are not currently executable. Invalid-action masking is theoretically compatible with policy-gradient optimization and becomes increasingly useful as the number of invalid actions grows [16].

III. Reinforcement Learning formulation and System substrate: Environment, Observation, and Action Contracts

A. Environment and connection RL-SabanaHerons wiring.

Policy training was performed in SimRobot, which reproduces the rigid-body dynamics, contacts, and falls of the NAO V6. The simulated measurements are processed by the perception and cognition modules of the SabanaHerons stack and exposed to the Python environment at each cognition frame. As summarized in Figure 1, the RL agent selects the tactical skill and target, while the existing stack remains responsible for its physical execution. The bridge and environment implementation are available in the team's public code release [17], Figure 1 represents the wiring block for the traning loop.

B. Observation Space

The observation space covers 47 different features including both single-agent and multi-agent information, as detailed in Table I. Rather than representing a collection of independent features, the observation vector is treated as an interface contract between the environment and the policy.

TABLE I
ORGANIZATION OF THE OBSERVATION SPACE.

Observation Group	Index	Observation Group	Index
Robot self-state	0–2	Teammate relative slots	26–31
Ball relative state	3–8	Team ball	32–34
Ball perception freshness	9–12	Teammate engagement	35–37
Scoring affordances	13–15	Team-ball consensus	38
Passing affordance	16	Opponent goal vector	39–40
Nearby agent distances	17–22	Extended gate state	41–43
Legacy gate state	23–25	Tactical role one-hot	44–46

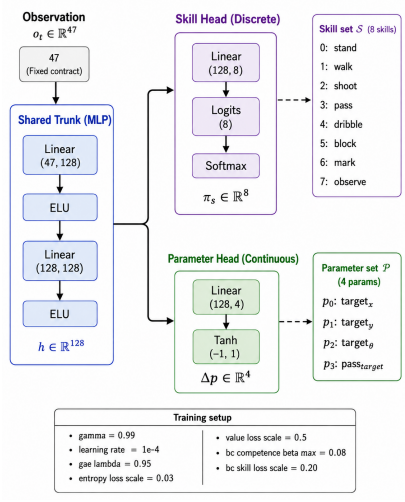


Fig. 3. Neural Network Architecture proposed for the agent.

C. Inter-Robot Communication

Inter-robot communication is limited by competition rules and by the onboard CPU, which must also handle perception, localization, and behavior; messages travel as UDP packets over a shared WiFi network and are frequently lost. Team messages therefore carry only a few features: robot pose and its quality, local ball state, behavior target, and strategic role. The complete message structure is available in [17].

D. Neural Network Architecture

The actor is a compact multilayer perceptron with ELU activations and a shared trunk, as shown in Figure 3, small enough to run on the NAO V6. Two heads branch from the trunk: a skill head that produces logits over the discrete skill set $\mathcal{S}$, and a parameter head that produces bounded residual parameters through a `tanh` activation. The network therefore operates at a high cognition level, specifying which skill to execute and its target rather than commanding motors directly.

E. Action Space

The hybrid action space combines discrete skill selection with continuous, role and skill dependent target parameters; unused parameters are masked. It also defines the interface with the SabanaHerons/B-Human execution stack: role- and state-conditioned masks remove actions incompatible with the assigned role or current gate state before selection, and the selected skill and decoded residuals are converted into an executable `SkillRequest`.

TABLE II
STAGED TRAINING CURRICULUM FROM SCENARIO-GUIDED REINFORCEMENT LEARNING TO MATCH-BASED POLICY OPTIMIZATION.

Training Stage	Tactical Content and Learning Strategy
Individual skills (v1)	Approach, takeover, dribble and shoot; BC + PPO
Multiagent attack (v2b)	Role-conditioned team attack; multi-agent PPO with team rewards
Multiagent cooperation (v3–v4.2)	Passing and offensive coordination; teacher-guided BC + KL-anchored PPO
Defensive training (Stage 8)	Goal-side positioning, marking and blocking
Unified policy (v5)	Attack–defence integration; BC + PPO fine-tuning with KL anchoring
Striker adaptation (DAgger)	Expert correction on live match states against reactive B-Human
Multiagent distillation	Progressive support-role handover using B-Human shadow labels
Learned coordination (v0–v2)	Neural support positioning learned from B-Human target positions
Match-based PPO (iter. 0–12)	On-policy match optimization with decaying teacher CE and KL anchoring

IV. FROM HEURISTIC TEACHER TO REINFORCEMENT-LEARNED FOOTBALL

Humanoid football combines sparse, delayed objectives with skill availability gated by geometric preconditions. Under this structure, PPO training alone can converge to a locally stable but tactically incomplete policy. Figure 8 in Section VI compares skill selection across training configurations. The principal configurations were evaluated over six random seeds, whereas the extended PPO-only baseline at 30k steps used three seeds. PPO-only training predominantly selected walking, whereas teacher-guided configurations retained more diverse skill selection. The agent overfited to walk rather to learn to use other actions from the action space in comparison with teacher based frameworks which used a more varied action space. The PPO agent passed $0/5$ of the production scenarios defined in [17], despite entering the shoot band on 10% of steps, motivating the teacher-student design of Section IV-B.

A. Curriculum learning

Dataset generation and training were organized into the stages of Table II, which decompose the task into objectives of increasing complexity without changing the structure of the observation and action spaces. Stages from individual skills -v1- to unified policy -v5- teacher as training agents were on static conditions, from striker adaptation to the final agent was in dynamic matches against live Bhuman. Stage definitions and promotion criteria are documented in [17].

B. Teacher-student

During scenario-based training, a fixed, role-specific heuristic teacher mapped local game geometry into tactical skill decisions, as summarized in Table III. In subsequent match-based

TABLE III
TACTICAL DECISION LOGIC AND EVOLUTION OF SUPERVISION THROUGHOUT TRAINING.

Tactical Function	**Scenario-Based Teacher**	**Match-Based Learning**
Role assignment	Ball proximity, possession override, and temporal hysteresis.	Deterministic assignment retained; role-conditioned policy.
Ball acquisition	Ball far → approach; close and aligned → takeover.	Striker DAgger using expert labels on student-visited states.
Finishing	Open shooting opportunity → shoot.	Expert supervision incorporating B-Human's kick macro.
Support positioning	Select open targets maintaining teammate separation; otherwise hold.	B-Human shadow targets; supervised neural coordinator.
Defensive coverage	Maintain goal-side positioning complementing goalkeeper coverage.	B-Human support decisions distilled from live matches.
Blocking and marking	Ball in front → block; nearby opponent → goal-side marking.	Positional defence learned from B-Human; block/mark not selected in final matches.
Recovery and clearance	Deep recovery → safe pass, clearance, or escape dribble.	Role-specific live supervision, followed by match-based PPO.

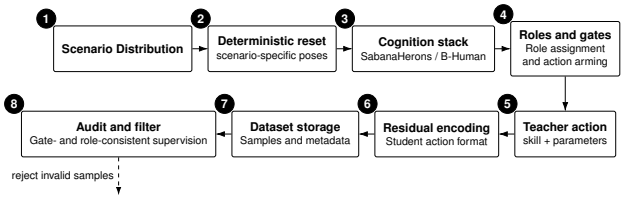


Fig. 4. Teacher-data generation and supervision-auditing pipeline.

training, DAgger collected student-visited states labeled by the expert for the striker, while support behaviors were distilled from B-Human's own decisions. The resulting policies were further optimized through on-policy PPO against reactive B-Human opponents. Separately, a neural positioning coordinator was trained by supervised regression on B-Human's support targets, replacing the heuristic positioning anchor while preserving deterministic role assignment.

Since 96.6% of B-Human's support targets exceeded the policy's residual positioning range, a separate MLP was trained through supervised regression to predict support targets directly from the 47-dimensional observation.

C. Dataset generation and auditing

Deterministic teacher policies generate the dataset through the pipeline shown in Figure 4, starting from scenario-specific resets of the robots and the ball. At each step, the tactical role and skill gates are written into the observation, and the teacher action is encoded into the same residual parameterization used by the student, keeping supervised labels aligned with the agent output. Samples inconsistent with the scenario, role, or gate state are discarded, as are episodes with sustained teammate collisions or backend failures.The generation and auditing pipeline is documented in [17], and the per-stage dataset summary is available in [18].

D. Behavioral cloning warm start

A student policy is first trained on the generated dataset through behavioral cloning, turning the deterministic observation-action mapping into an initial set of neural parameters. Training follows a strict checkpoint lineage: at each tuning stage, a validated model is loaded as the warm start of the new trainable policy, and a frozen copy serves as the KL anchor during PPO.

E. KL anchor, BC replay and Online Policy Training

Policy training combined expert supervision with reinforcement learning to limit drifts from previously acquired tactical behaviors. The discrete skill component was trained using cross-entropy:

$$\mathcal{L}_{\mathrm{CE}} = -\mathbb{E}_{(x,s^*)\sim\mathcal{D}}\left[\log \pi_\theta(s^* \mid x)\right], \tag{1}$$

where x denotes the observation, s^* the expert skill label, and $\mathcal{D}$ the corresponding supervised dataset. During scenario-based PPO, continuous action residuals were additionally supervised through Smooth L1 regression. This residual regression term was not used in the subsequent match-based PPO stage.

A second regularization mechanism constrained deviations from a frozen reference policy using KL divergence:

$$\begin{aligned}\mathcal{L}_{\mathrm{KL}} &= D_{\mathrm{KL}}\left(\pi_\theta(\cdot \mid x) \| \pi_{\mathrm{ref}}(\cdot \mid x)\right) \\ &= \sum_{s\in\mathcal{S}(x)} \pi_\theta(s \mid x) \log \frac{\pi_\theta(s \mid x)}{\pi_{\mathrm{ref}}(s \mid x)}.\end{aligned} \tag{2}$$

$\mathcal{S}(x)$ denotes the legal skill set under the action mask. Initially, the reference corresponded to a previously trained checkpoint, aiming to limit behavioral drift during attack–defence integration. During match-based optimization, the reference was replaced by the distilled imitation policy, providing an anchor to expert-derived behavior.

Following DAgger-based striker adaptation and the distillation of B-Human support decisions, the policy was further optimized through on-policy PPO in 4v4 SimRobot matches against reactive B-Human opponents. The objective was defined as

$$\begin{aligned}\mathcal{L}_{\mathrm{clip}} = -\,\mathbb{E}_t\Big[\min\big(&r_t(\theta)\hat{A}_t, \\ &\mathrm{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\big)\Big],\end{aligned} \tag{3}$$

where

$$r_t(\theta) = \frac{\pi_\theta(a_t \mid x_t)}{\pi_{\theta_{\mathrm{old}}}(a_t \mid x_t)}, \tag{4}$$

and $\hat{A}_t$ is the advantage estimate computed using Generalized Advantage Estimation:

$$\begin{aligned}\delta_t &= r_t^{\mathrm{env}} + \gamma V_\phi(x_{t+1}) - V_\phi(x_t), \\ \hat{A}_t &= \sum_{l=0}^{T-t-1} (\gamma\lambda_{\mathrm{GAE}})^l \delta_{t+l}.\end{aligned} \tag{5}$$

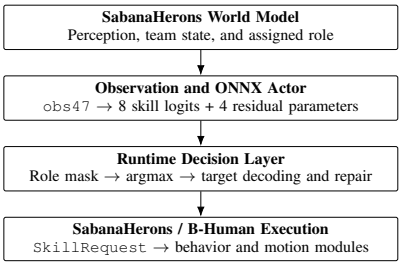


Fig. 5. Deployment pipeline diagram of the final ONNX agent.

Match-based PPO used $\gamma = 0.995$, $\lambda_{\text{GAE}} = 0.95$, and $\epsilon = 0.2$. Trajectories were collected from simulated matches and optimized through minibatch updates.

The regularized training objectives differed between the two stages. During scenario-based PPO,

$$\mathcal{L}_{\text{scenario}} = \mathcal{L}_{\text{PPO}} + \beta_{\text{BC}}\mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{res}} + \lambda^{A}_{\text{KL}}\mathcal{L}^{A}_{\text{KL}}, \quad (6)$$

where $\mathcal{L}_{\text{res}}$ represents the weighted Smooth L1 residual regression term, and β_{BC} decreased from 0.08 to 0.02 during the attack–defence fine-tuning stage.

During match-based PPO, the objective became

$$\mathcal{L}_{\text{match}} = \mathcal{L}_{\text{PPO}} + \beta_{\text{T}}\mathcal{L}_{\text{CE}} + \lambda^{B}_{\text{KL}}\mathcal{L}^{B}_{\text{KL}}, \quad (7)$$

where the teacher supervision coefficient β_{T} decreased from 0.30 to 0.004, and the KL reference was the frozen round-3 distilled policy. Thus, match-based optimization retained a supervised skill regularizer while removing the Smooth L1 residual regression term.

V. Deployment Pipeline For The Final ONNX Agent

During deployment, the final policy runs in the SabanaHerons cognition loop of each field player, as shown in Figure 5. At each frame, the robot's world model, the shared team state, and the assigned role are encoded into the same 47-dimensional observation used during training. The actor returns eight skill logits and four bounded parameters; a role-conditioned mask removes skills invalid for the current role and game state, and an argmax selects the highest-valued remaining skill.

The resulting target is constrained to a valid field position and converted, together with the selected skill, into a native `SkillRequest` executed by the existing SabanaHerons modules. Roles are committed beforehand by a deterministic coordinator, whose rules and constants are available in [17].

VI. Experimentation

The proposed pipeline was evaluated through complementary criteria: performance in controlled tactical scenarios, agreement with expert skill decisions, preservation of acquired competencies during PPO fine-tuning, and competitive performance in multi-agent matches. This evaluation distinguishes policy optimization from behavioral competence, since episodic return alone does not establish whether an agent retains the tactical skills required for football.

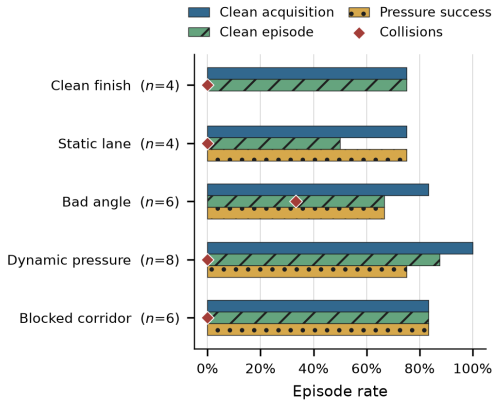


Fig. 6. Agent performance across five production scenarios. Clean acquisition indicates successful ball takeover; pressure success represents obstacle clearance; diamonds indicate collision rate.

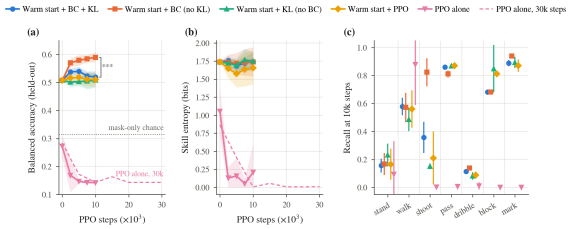


Fig. 7. Ablations on each main part of the training pipeline and their contribution to skill logits.

Five production scenarios were evaluated across 28 SimRobot episodes, including ball acquisition, static-lane navigation, adverse shooting angles, dynamic pressure, and blocked corridors. Figure 6 reports task success and collision outcomes. Skill-level evaluations were conducted on held-out expert-labeled observations, while the principal training ablations used six random seeds. An extended PPO-only baseline was additionally evaluated up to 30k steps.

Competitive performance was assessed through 79 side-balanced, ten-minute 4v4 SimRobot matches against reactive B-Human opponents, comprising 20 imitation-only, 28 imitation-plus-PPO, and 31 stock B-Human reference matches.

A. Results and Discussion

The initial design hypothesis was that BC replay and KL regularization would preserve teacher-acquired competencies during the pipeline. However, the ablations shows the opposite effect from KL. As shown in Figure 7 in the part a, removing KL while retaining BC increased held-out balanced accuracy from approximately 0.52 to 0.59. Meanwhile, the warm-start PPO configuration without either auxiliary regularizer achieved approximately 0.51, suggesting that initialization itself contributed substantially to retaining expert-aligned behavior.

The PPO-only baseline exhibited a markedly different behavior. Figures 8 and 7shows that its balanced accuracy de-

TABLE IV
MATCH PERFORMANCE OF IMITATION-ONLY, IMITATION WITH PPO, AND STOCK B-HUMAN AGAINST B-HUMAN OPPONENTS IN 4V4 SIMROBOT.

Metric	Imitation-only (DAgger + Distillation)	Imitation + PPO (Final Policy)	Stock B-Human (Reference)
Matches	20	28	31
W–D–L	6–0–14	7–4–17	15–4–12
Goal difference	-1.15	-1.07	$+0.52$
95% CI (GD)	$[-2.30, 0.05]$	$[-2.00, -0.14]$	$[-0.52, 1.52]$
Goals for	1.20	1.21	2.48
Goals against	2.35	2.1	1.97
Possession (%)	33.3	34.9	29.8
Ball control (%)	17.4	19.1	15.5
Opponent half (%)	22.8	24.6	30.2
Final third (%)	4.6	4.3	13.6
Shots on target	3.0	3.3	4.5
Passes	6.2	7.4	8.7
Recoveries	29.0	31.1	22.4
Duels won (%)	30.2	31.2	26.9
Turnovers	28.8	31.0	22.5
Falls	15.1	15.4	20.1

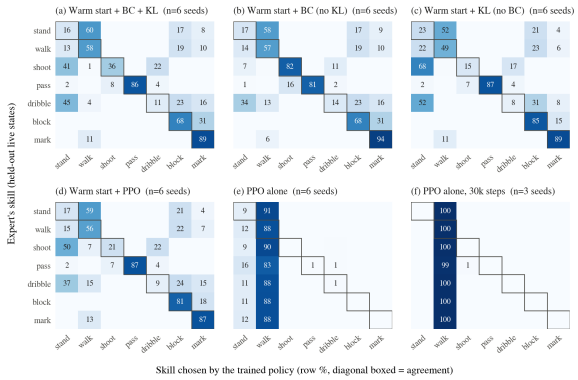


Fig. 8. Skill confusion matrix from the trained agents in contrast with the teacher.

creased toward 0.15, while skill-selection entropy approached zero. The corresponding confusion matrix shows a strong concentration on walking, indicating that reward optimization without teacher-derived initialization failed to maintain a diverse expert-aligned skill selection. The warm-start configurations retained broader skill distributions and substantially higher agreement with expert decisions.

Together, these results demonstrate that comparable optimization progress does not necessarily imply preservation of tactical competencies.

Table IV reports the competitive performance of the final imitation-plus-PPO policy, its imitation-only predecessor, and stock B-Human. The final policy achieved higher ball possession, ball control, and recoveries per match than the B-Human reference. It also recorded fewer falls, suggesting favorable behavior in several interaction and recovery metrics.

The final policy achieved a mean goal -1.07, compared with -1.15 for imitation-only and $+0.52$ for stock B-Human. Moreover, the final policy spent only 4.3% of ball time in the attacking third, compared with 13.6% for B-Human. This discrepancy indicates that increased possession and recovery activity did not translate into comparable territorial progression or scoring performance.

The experiments support the feasibility of integrating teacher-guided learning with deterministic role assignment and role-conditioned action constraints. However, they also expose three important limitations: the dependence of knowledge transfer on expert demonstration coverage, the absence of an additional accuracy benefit from KL regularization in the evaluated ablations, and the discrepancy between skill-level competence and competitive match performance.

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